

OXFORD

INTERNATIONAL
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INTERNATIONAL GCSE **CHEMISTRY**

9202/1

Paper 1

Mark scheme

June 2023

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Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from oxfordaqaexams.org.uk

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly Level 3 with a small amount of Level 4 material it would be placed in Level 3 but be awarded a mark near the top of the level because of the Level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives, level of demand and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available ‘any **two** from’ is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that ‘right + wrong = wrong’.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(...) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

'Ignore' is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

'Do **not** accept' means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	cracking		1	AO1 3.10.1.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	C ₂ H ₄		1	AO2 3.10.1.3c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	a double (covalent) bond		1	AO1 3.10.1.3c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	(test) bromine (water)		1	AO1 3.10.1.3d
	(result bromine water) turns from orange to colourless	allow (bromine water) is decolourised MP2 dependent on MP1 being awarded	1	3.10.1.3b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	poly(ethene)	allow polyethene / polythene	1	AO1 3.10.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.6	biodegradable	allow broken down by microbes	1	AO1 3.10.2e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.7	any one from: • recycle • reuse • use less polymers • use biodegradable polymers • burn polymers		1	AO2 3.10.2e

Total Question 1		8
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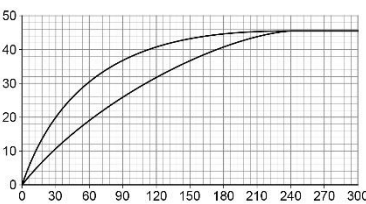
Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	concentration of hydrochloric acid		1	AO4 3.8.1f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	26 (cm ³)		1	AO2 3.8.1f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	(rate of reaction =) $\frac{43}{200}$		1	AO2 3.8.1a
	(rate of reaction =) 0.215 (cm ³ /s)		1	

Question	Answers	Extra information*	Mark	AO/ Spec. Ref.
02.4	smaller pieces of calcium carbonate have a greater surface area		1	AO1 3.8.1f
	there are more particle collisions per second		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	line starting at 0,0 with steeper increase levels off at the same level	allow a tolerance of $\pm \frac{1}{2}$ a small square an answer of  scores 2 marks	1 1	AO3 3.8.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.6	any one from: - change of concentration of hydrochloric acid - change of temperature (of hydrochloric acid)	allow increase / decrease for change of allow use of a catalyst	1	AO2 3.8.1c 3.8.1e

Total Question 2		9
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Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	sports injury packs		1	AO1 3.9.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	(as the mass increases the temperature) decreases		1	AO3 3.9.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	line extrapolated to 6 g 12 (°C)	allow correct reading from extrapolated line allow tolerance of $\pm \frac{1}{2}$ a small square	1	AO3 3.9.2c
			1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	any one from: • use a plastic / polymer beaker / cup • lagging / insulation • add a lid.		1	AO3 3.9.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.5	(temperature change =) 5.5 (°C)		1	AO2 3.9.2a
	(Q =) $50 \times 4.2 \times 5.5$	allow correct use of an incorrectly determined temperature change	1	
	= 1155 (J)	ignore sign	1	
Total Question 3			8	

Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	incomplete combustion (in Equation 2)		1	AO2 AO3 3.10.1.2d
	(because) less / insufficient oxygen is supplied		1	
	OR			
	complete combustion (in Equation 1) (1)			
	(because) more / enough oxygen is supplied (1)			

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	global dimming		1	AO1 3.10.1.2d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	(there is) sulfur (in fuel)	allow sulfur combines with oxygen	1	AO1 3.10.1.2d
	(the sulfur) burns		1	
	to produce sulfur dioxide		1	
	OR			
	nitrogen combines with oxygen (1)			
	at high temperature (1)			
	to produce oxides of nitrogen (1)			

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	Level 3: A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.		5–6	AO1 AO2 AO3 3.9.3b 3.10.1.2d
	Level 2: Some logically linked reasons are given. There may also be a simple judgement.		3–4	
	Level 1: Relevant points are made. They are not logically linked.		1–2	
	No relevant content.		0	
	Indicative content • diesel comes from crude oil which is non-renewable • diesel produces carbon dioxide that contributes to climate change • diesel produces particulates that contributes to global dimming • hydrogen is produced from methane which is non-renewable • hydrogen only produces water so no pollutants are produced • hydrogen is a gas so is more difficult to store • hydrogen is a gas so takes up more space than diesel • hydrogen gas is explosive • hydrogen gas is not readily available • more energy is produced per gram of fuel from hydrogen so a smaller mass of fuel is used • reasoned judgement			

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.5	(M_r of decane $\Rightarrow$) 142		1	AO2 3.9.2b
	$\left(\text{mol of decane} = \frac{0.426}{142} \right)$ 0.003 (mol)	allow correct use of incorrectly determined value of M_r	1	
	(unit conversion $= \frac{19\,500}{1000} \Rightarrow$) 19.5 (kJ)		1	
	(heat energy change per mol $\Rightarrow$) $\frac{19.5}{0.003}$	allow correct use of no / incorrect unit conversion	1	
	$= (-) 6\,500$ (kJ)		1	
	Alternative approach			
	(M_r of decane $\Rightarrow$) 142 (1)			
	$\left(\text{mol of decane} = \frac{0.426}{142} \right)$ 0.003 (mol) (1)	allow correct use of incorrectly determined value of M_r		
	(heat energy change per mol $\Rightarrow$) $\frac{19\,500}{0.003}$ (1)			
	$= (-) 6\,500\,000$ (1)			
	$= (-) 6\,500$ (kJ) (1)	allow correct conversion of incorrectly determined value of heat energy change per mole		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.6	energy must be supplied to break bonds		1	AO1 3.9.2e 3.9.2f
	(and) energy is released when bonds are formed		1	
	(so) the energy needed to break existing bonds is less than the energy released to form bonds		1	
Total Question 4			20	

Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	zinc carbonate → zinc oxide + carbon dioxide		1	AO2 3.3.1.2a 3.6.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	(test) (bubble the gas through) limewater	allow calcium hydroxide solution for limewater	1	AO1 3.4.2c 3.4.3d
	(result) (limewater) turns cloudy (white)	allow (limewater) turns milky MP2 dependent on MP1 being awarded	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	(M_r of ZnCO_3 =) 125	allow correct use of incorrectly determined M_r	1	AO2 3.6.2b
	(percentage by mass =) $\frac{65}{125} \times 100$		1	
	= 52 (%)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.4	3.41 (g)		1	AO2 3.6.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.5	mass would stop decreasing (as) reaction is complete	allow no more carbon dioxide given off	1	AO3 3.3.1.2a 3.6.1a
	OR mass would (continue to) decrease (1) (as) zinc carbonate had not fully decomposed after 280 s (1)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.6	(volume =) 0.079×24	allow correct conversion of an incorrect determined value of molar volume	1	AO2 3.6.4d
	= 1.896 (dm ³)		1	
	(unit conversion =) 1896 (cm ³)		1	
	= 1900 (cm ³)		1	
	alternative approach	allow an answer correctly rounded to 2 significant figures from an incorrect calculation of molar volume using data from the question		
	(unit conversion = 24 × 1000 =) 24000 (cm ³) (1)			
	(volume =) 0.079×24000 (cm ³) (1)			
	= 1896 (cm ³) (1)			
	= 1900 (cm ³) (1)	allow correct use of incorrect / no conversion		
		allow an answer correctly rounded to 2 significant figures from an incorrect calculation of molar volume using data from the question		

Total Question 5		13
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Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	titanium		1	AO2 3.1.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	conducts heat		1	AO1 3.3.1c
	strong		1	3.7.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	a substance made up of only one sort / type of atom		1	AO1 3.1.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	(transition metal compounds) are coloured		1	AO3 3.7.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.5	iron	allow Fe	1	AO1 3.8.3c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.6	(catalysts) provide a different pathway for the reaction		1	AO1 3.9.2h
	(which has a) lower activation energy		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.7	(cobalt) forms ions with different charges		1	AO2 3.7.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.8	strong electrostatic forces of attraction		1	AO1 3.2.1j
	(between) positive ions and electrons		1	
	(outer) electrons are delocalised		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.9	$\text{Cu} + 2 \text{Ag}^+ \rightarrow \text{Cu}^{2+} + 2 \text{Ag}$	allow $\text{Cu} + \text{Ag}^+ \rightarrow \text{Cu}^{2+} + \text{Ag}$ for 1 mark	2	AO2 3.3.1.1b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.10	(because) silver ions gain electrons	do not accept references to loss of oxygen allow gain of electrons for 1 mark	2	AO2 3.3.2d

Total Question 6		16
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Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	homologous (series)		1	AO1 3.10.3.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	hydrogen		1	AO1 3.10.3.1b

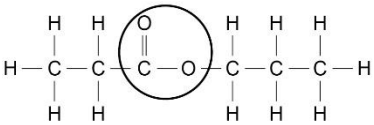
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	7		1	AO1 3.10.3.1b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.4	$C_6H_{12}O_6 \rightarrow 2 C_2H_5OH + 2 CO_2$	allow 1 mark for $C_2H_5OH + CO_2$	2	AO1 3.10.1.2f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.5	any two from: • faster • more efficient • continuous process • less labour intensive • product is purer • does not use crops that could be used as food	allow not batch process	2	AO2 3.10.1.2f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.6	oxidation		1	AO1 3.10.3.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.7	intermolecular forces	allow forces between the molecules	1	AO2 AO3 3.2.2d 3.10.1.2c
	(are) stronger (the higher the number of carbon atoms)		1	
	(so) more energy is required to overcome the forces		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.8			1	AO2 3.10.3.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.9	(alcohol) propanol		1	AO2 3.10.3.3a
	(carboxylic acid) propanoic acid		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.10		answers may be in either order		AO1 3.10.3.3b
	distinctive / pleasant smell		1	
	volatile		1	

Total Question 7		16
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